Halward Inspection Services Ltd.

Inspection Report

Site: Hamilton Plant

No.	Status	Date
HIS-2011-0207-DS-6-RA-103	Issued	2011-02-18

UT Thickness Data Sheet - 6-RA-103

Scope and Responsibilities

This data sheet provides the ultrasonic thickness measurement results for line 6-RA-103, a rich amine header located within the ATU-100. The inspection was conducted to support the 5-year Class 1 survey as detailed in HIS-2011-0207.

Kevin Doucette, acting as lead UT technician (SNT-TC-1A UT Level II), performed the measurements and data recording. All activities were executed on behalf of Halward Inspection Services Ltd. to ensure asset integrity for the Hamilton Plant.

Technical Specifications

The subject line, 6-RA-103, is constructed from A106 Gr.B with an outside diameter of 168.3 mm and a nominal thickness of 10.97 mm. The design limits for this service are 862 kPa(g) and 82 °C.

The minimum allowable thickness (t-min) is established at 4.57 mm. Any measured value falling below this threshold is subject to an NCR in accordance with site retirement thickness rules.

Procedural Compliance

Thickness measurements were performed under the requirements of QSCP-NDT-001 Rev 2. This procedure incorporates specific CUI program mandates, including insulation removal, reinstatement, and the use of inspection ports and profile radiography.

Field execution followed the methodology outlined in HIS-UT-002 Rev 1, which was submitted by Halward Inspection Services Ltd. for QSCP acceptance. All ultrasonic testing was conducted in accordance with ASTM E797.

Analysis and Interval Determination

Based on the data recorded in HIS-2011-0207, the rich amine service in ATU-100 is currently at half remaining life. The corrosion rate was determined by comparing the long-term rate against the short-term rate from the latest pair of readings taken at least 3 years apart.

The next thickness measurement interval is set based on the remaining life calculation (t - 4.57 mm) / rate. This interval shall not exceed the Class 1 maximum of 5 years.

Thickness readings, measured 2011-02-18

Line 6-SG-101 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-101-01	7.11	3.56	2006-02-24	6.59	6.39	0.049	0.040	>50	2016-02-18	
6-SG-101-02	7.11	3.56	2006-02-24	6.51	6.14	0.061	0.074	34.7	2016-02-18	
6-SG-101-03	7.11	3.56	2006-02-24	6.42	6.17	0.061	0.050	42.9	2016-02-18	
6-SG-101-04	7.11	3.56	2006-02-24	6.28	6.05	0.070	0.046	35.8	2016-02-18	
6-SG-101-05	7.11	3.56	2006-02-24	6.22	5.87	0.085	0.070	27.2	2016-02-18	
6-SG-101-06	7.11	3.56	2006-02-24	6.49	6.27	0.054	0.044	>50	2016-02-18	
6-SG-101-08	7.11	3.56	—	—	6.33	—	—	—	2016-02-18	

Line 6-SG-102 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-102-01	7.11	3.56	2006-02-24	6.26	5.75	0.093	0.102	21.4	2016-02-18	
6-SG-102-02	7.11	3.56	2006-02-24	6.25	5.80	0.088	0.090	24.8	2016-02-18	
6-SG-102-03	7.11	3.56	2006-02-24	6.26	5.80	0.083	0.092	24.3	2016-02-18	
6-SG-102-04	7.11	3.56	2006-02-24	6.22	5.73	0.084	0.098	22.1	2016-02-18	
6-SG-102-05	7.11	3.56	2006-02-24	6.72	6.46	0.045	0.052	>50	2016-02-18	
6-SG-102-06	7.11	3.56	—	—	6.41	—	—	—	2016-02-18	

Line 4-RA-101 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-101-01	8.56	3.81	2006-02-24	8.27	8.32	0.013	-0.010	>50	2021-02-18	
4-RA-101-02	8.56	3.81	2006-02-24	8.27	8.20	0.023	0.014	>50	2021-02-18	
4-RA-101-03	8.56	3.81	2006-02-24	8.10	7.98	0.042	0.024	>50	2021-02-18	
4-RA-101-04	8.56	3.81	2006-02-24	8.34	8.21	0.028	0.026	>50	2021-02-18	
4-RA-101-05	8.56	3.81	2006-02-24	8.29	8.16	0.029	0.026	>50	2021-02-18	
4-RA-101-07	8.56	3.81	2006-02-24	8.30	8.30	0.020	0.000	>50	2021-02-18	
4-RA-101-08	8.56	3.81	2006-02-24	8.07	8.02	0.044	0.010	>50	2021-02-18	
4-RA-101-09	8.56	3.81	2006-02-24	8.43	8.32	0.017	0.022	>50	2021-02-18	
4-RA-101-10	8.56	3.81	2006-02-24	8.27	8.28	0.026	-0.002	>50	2021-02-18	
4-RA-101-11	8.56	3.81	2006-02-24	8.14	8.04	0.043	0.020	>50	2021-02-18	

Line 4-RA-102 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-102-01	8.56	3.81	2006-02-24	8.41	8.33	0.014	0.016	>50	2021-02-18	
4-RA-102-02	8.56	3.81	2006-02-24	8.15	7.99	0.039	0.032	>50	2021-02-18	
4-RA-102-03	8.56	3.81	2006-02-24	8.27	8.04	0.035	0.046	>50	2021-02-18	
4-RA-102-04	8.56	3.81	2006-02-24	8.24	7.96	0.039	0.056	>50	2021-02-18	
4-RA-102-05	8.56	3.81	2006-02-24	8.19	8.07	0.032	0.024	>50	2021-02-18	
4-RA-102-06	8.56	3.81	2006-02-24	8.15	8.01	0.033	0.028	>50	2021-02-18	
4-RA-102-07	8.56	3.81	2006-02-24	8.29	8.10	0.032	0.038	>50	2021-02-18	
4-RA-102-08	8.56	3.81	2006-02-24	8.05	7.88	0.041	0.034	>50	2021-02-18	
4-RA-102-09	8.56	3.81	2006-02-24	8.18	7.98	0.038	0.040	>50	2021-02-18	
4-RA-102-11	8.56	3.81	—	—	8.19	—	—	—	2021-02-18	

Line 6-RA-103 | Rich amine header | A106 Gr.B | t_nom 10.97 mm | t-min 4.57 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-RA-103-01	10.97	4.57	2006-02-24	10.76	10.47	0.035	0.058	>50	2021-02-18	
6-RA-103-02	10.97	4.57	2006-02-24	10.50	10.23	0.058	0.054	>50	2021-02-18	
6-RA-103-03	10.97	4.57	2006-02-24	10.67	10.42	0.038	0.050	>50	2021-02-18	
6-RA-103-04	10.97	4.57	2006-02-24	10.40	9.97	0.068	0.086	>50	2021-02-18	
6-RA-103-05	10.97	4.57	2006-02-24	10.34	10.10	0.057	0.048	>50	2021-02-18	
6-RA-103-06	10.97	4.57	2006-02-24	10.74	10.56	0.030	0.036	>50	2021-02-18	
6-RA-103-07	10.97	4.57	2006-02-24	10.66	10.55	0.029	0.022	>50	2021-02-18	
6-RA-103-08	10.97	4.57	2006-02-24	10.58	10.44	0.035	0.028	>50	2021-02-18	
6-RA-103-09	10.97	4.57	2006-02-24	10.67	10.56	0.030	0.022	>50	2021-02-18	
6-RA-103-10	10.97	4.57	2006-02-24	10.27	10.05	0.065	0.044	>50	2021-02-18	
6-RA-103-13	10.97	4.57	—	—	10.45	—	—	—	2021-02-18	

Line 4-RA-104 | L/R exchanger, rich side | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-104-01	8.56	3.81	2006-02-24	8.16	7.95	0.037	0.042	>50	2021-02-18	
4-RA-104-02	8.56	3.81	2006-02-24	8.23	8.14	0.027	0.018	>50	2021-02-18	
4-RA-104-03	8.56	3.81	2010-09-13	8.59	8.56	0.069	0.069	>50	2021-02-18	
4-RA-104-04	8.56	3.81	2010-09-13	8.52	8.60	-0.185	-0.185	>50	2021-02-18	
4-RA-104-05	8.56	3.81	2010-09-13	8.43	8.57	-0.324	-0.324	>50	2021-02-18	
4-RA-104-06	8.56	3.81	2006-02-24	8.09	7.98	0.041	0.022	>50	2021-02-18	
4-RA-104-07	8.56	3.81	2006-02-24	7.97	7.78	0.054	0.038	>50	2021-02-18	
4-RA-104-08	8.56	3.81	—	—	8.18	—	—	—	2021-02-18	

Sign-off	Name	Function	Date	Signature
Prepared	Kevin Doucette	lead UT technician, SNT-TC-1A UT Level II	2011-02-18	_____